

Total No. of Questions : 8]

SEAT No. :

PC-2499

[Total No. of Pages : 2

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B.E.(Mechanical)

ENERGY AUDIT AND MANAGEMENT

(2019 Pattern) (Semester - VIII) (402050B) (Elective - V)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Use of calculator is allowed.
- 4) Assume suitable data if necessary.

Q1) a) What is sensitivity and risk analysis? Explain the factors affecting. [9]

- b) Investment for an energy proposal is Rs.10.00 lakhs. Annual savings for the first three years is 150,000, 200,000 & 300,000. Considering cost of capital as 10%, what is the net present value of the proposal? [8]

OR

Q2) a) Write Notes on A) Return on Investment. B) Internal Rate of Return.[10]

- b) A sum of Rs.4, 00, 000/- is deposited in a bank at the beginning of a year. The bank pays 5% interest annually. How much money is in the bank account at the end of tenth year, if no money is withdrawn? [7]

Q3) a) What are different Energy Conservation Opportunities in Boiler System? [8]

- b) A centrifugal pump is pumping 85 m³/hr. of water and pressure rise in the pump is 6 bar (guage). If power drawn by motor is 25KW. Find out the pump efficiency. Assume motor efficiency as 90% & water density as 998 Kg/m³. [10]

OR

P.T.O.

Q4) a) What are the different Energy Conservation Opportunities in cooling tower and pumping system? [8]

b) Find out the efficiency of boiler by direct methods with the data given below: [10]

Type of boiler: oil fired

Quantity of steam generated (dry): 6000 kg/hr

Steam pressure gauges/ Temp: 10 bar /180 degree Celsius

Quantity of coal consumed: 410 kg/hr

Feed water temperature: 70 degrees Celsius and C_{pf} is 4.2

GCV of fuel: 43500 kJ/kg

Enthalpy of saturated steam at 10 bar pressure :2776 kJ/kg

Enthalpy of feed water: 70 kJ/kg

Q5) a) Explain various energy saving opportunities in electrical system? [7]

b) The lighting connected load for the small industry consisting of 140 Fluorescent tubes of 55 W each with magnetic ballast. In first option, the magnetic ballast of fluorescent tubes is replaced by electronic ballast & power consumption of same fluorescent tubes reduces to 40W. Calculate the simple payback period of above replacement if cost of electronic ballast is Rs. 110. In second option, fluorescent tubes are replaced by energy efficient fluorescent tubes of 20 W & cost of Rs. 450 each. Calculate simple payback period. Which energy saving option is better & why? Consider usage of 16 hrs per day & an electrical tariff of Rs. 4 per KWh. [10]

OR

Q6) a) Discuss how selection and location of transformer affect the power factor. [9]

b) Explain Energy Efficient Motors. How motor selection is done? [8]

Q7) a) Describe suitable factors influencing selection of cogeneration plan? [9]

b) Explain CDM project with flowchart. [9]

OR

Q8) a) Explain the term topping cycle and bottoming cycle with examples. [9]

b) What is the cogeneration? Describe technical option for cogeneration and write down advantages of cogeneration. [9]

